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[NPTEL \(https://swayam.gov.in/explorer?ncCode=NPTEL\)](https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Database Systems (course)

Course outline

[About NPTEL \(\)](#)[Introduction to Database Systems \(\)](#)[Week 0 \(\)](#)[Week 1 \(\)](#)[Week 2 \(\)](#)[Week 3 \(\)](#)

Week 3 : Assignment 3

The due date for submitting this assignment has passed.

Due on 2026-02-11, 23:59 IST.

Assignment submitted on 2026-02-05, 18:24 IST

1) A relation is a set of tuples.

0.5 points

☒ True

☐ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

True

☐ Relational Data Model and Notion of Keys (unit? unit=31&lesson=32)

☐ Introduction to Relational Algebra (unit? unit=31&lesson=33)

☐ Operators in Relational Model (unit? unit=31&lesson=34)

☐ Uses of Renaming, Join and Division in Relation Algebra (unit? unit=31&lesson=35)

☐ Lecture Slides (unit? unit=31&lesson=36)

☐ Week 3 Feedback Form: Introduction to Database Systems!! (unit? unit=31&lesson=38)

☒ **Quiz: Week 3 : Assignment 3 (assessment? name=192)**

Week 4 ()

Week 5 ()

Week 6 ()

2) Different tuples in a relation may have different numbers of components.

0.5 points

- ☐ True
☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False

3) Each candidate key of a relation is also a primary key.

0.5 points

- ☐ True
☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False

4) The logical schema represents the entity sets of the domain being modeled and their relationships.

0.5 points

- ☐ True
☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False

5) A relation always has a key.

0.5 points

- ☒ True
☐ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

True

[Week 7 \(\)](#)[Week 8 \(\)](#)[Week 9 \(\)](#)[Week 10 \(\)](#)[Week 11 \(\)](#)[Week 12 \(\)](#)[Download Videos \(\)](#)[Lecture Notes \(\)](#)[Problem Solving
Session - Jan 2026 \(\)](#)

6) A relation may have more than one key.

0.5 points

- ☒ True
☐ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

True

7) A Relation can have at most one foreign key.

0.5 points

- ☐ True
☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False

8) A foreign key in a relation is also a key for that relation.

0.5 points

- ☐ True
☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False

9) Consider the following statements:

2 points

S1: An attribute of a relation may have a domain consisting of elements that are lists.

S2: No two attributes of a relation can have the same domain.

Choose the correct option:

- ☐ S1: False; S2: False

- ☒ S1: False; S2: True
☐ S1: True; S2: False
☐ S1: True; S2: True

No, the answer is incorrect.

Score: 0

Accepted Answers:

S1: False; S2: False

10) Consider a relation $R(\underline{A_1}, A_2, A_3, A_4, A_5)$ where attribute A_1 is the *only key*. The maximum possible number of super **2 points** keys for the relation R will be:

- ☐ 1
☐ 4
☐ 5
☒ 16

Yes, the answer is correct.

Score: 2

Accepted Answers:

16

11) Suppose $A_1 A_5$ is the key for the relation $R(A_1, A_2, A_3, A_4, A_5)$. Consider the statements given below:

2 points

S1: A_5 may also be a key for the relation R.

S2: $A_1 A_4 A_5$ is a super key for the relation R.

S3: $A_1 A_4 A_5$ may also be a key for the relation R.

Choose the correct option:

- ☐ S1: True; S2: True; S3: False
☐ S1: True; S2: False; S3: False
☐ S1: False; S2: False; S3: True
☒ S1: False; S2: True; S3: False

Yes, the answer is correct.

Score: 2

Accepted Answers:

S1: False; S2: True; S3: False

12) Consider the following relations P(A,B,C), Q(A,B,D), and R(B,E) and the given instances. What is the number of the tuples in the result of the relational algebra expression given below: **2 points**

$$\Pi_A(\sigma_{Q.B=P.B \wedge Q.D > 1}(P \times Q)) - \Pi_A(\sigma_{P.B=R.B \wedge R.E=12}(P \times R))$$

P

A	B	C
1	3	6
1	3	7
2	4	7
2	5	8

Q

A	B	D
2	3	2
1	4	5
2	3	6
0	9	1

R

B	E
3	11
9	12
4	13
4	12

- ☐ 0
☒ 1
☐ 2
☐ 4

Yes, the answer is correct.

Score: 2

Accepted Answers:

1

13) Consider the following schema: **2 points**

Suppliers(sid , sname, address)Parts(pid , pname, color)Catalog(sid,pid , cost)

Which one of the following relation algebra(RA) expressions correctly represents the following query?

"Find the names of the suppliers who supply some red part"

☐

$\pi_{sname}(\pi_{sid}((\pi_{pid}(\sigma_{color='red'}(P\ parts)) * Catalog) * Suppliers))$

☒

$\pi_{sname}(\pi_{sid}((\pi_{pid}(\sigma_{color='red'}(P\ parts)) \times Catalog) * Suppliers))$

☐

$\pi_{sname}(\pi_{sid}((\pi_{pid}(\sigma_{color='red'}(P\ parts)) \times Suppliers) * Catalog))$

☐

$\pi_{sname}((\pi_{pid}(\sigma_{color='red'}(P\ parts)) \times \pi_{sid}(Suppliers)) * Catalog)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\pi_{sname}(\pi_{sid}((\pi_{pid}(\sigma_{color='red'}(P\ parts)) * Catalog) * Suppliers))$

14) Consider two relations r_1 and r_2 with schemas R_1 and R_2 , respectively. Here, $|R_1| = m$ and $|R_2| = n$ and $m \geq 2$, $n \geq 2$ **2 points**
 2. Relation r_3 is the natural join of r_1 and r_2 ; i.e. $r_3 = r_1 * r_2$. Suppose r_1 and r_2 share k number of common attributes, where $k > 0$ and $k < \min(m, n)$. Identify the **FALSE** statement:

☒

The number of attributes in r_3 will be less than $m+n$.

☐

The minimum number of attributes in r_3 is: $\max(m, n)$.

☐

The scheme of r_3 will be: $R_1 \cup (R_2 - \{a_1, a_2, \dots, a_k\})$, where a_1 to a_k are common attributes.

☐

The maximum possible number of attributes in r_3 is: $m+n-1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The minimum number of attributes in r_3 is: $\max(m, n)$.

15) Consider a relation $P(A, B, C, D)$ and the given data instance. What is the result of the relational algebra expression $P \div Q$ if Q is computed as follows: **2 points**

$$Q = \Pi_{B,D}(\sigma_{C>5}(P))$$

P

A	B	C	D
1	5	3	2
2	5	6	2
6	2	2	5
1	5	1	2
1	2	1	5

1	2	3	5
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- ☐ A relation with scheme (A,C) and tuple (1,1)
☐ A relation with scheme (A,C) and tuple (1,3)
☒ A relation with scheme (A,C) and tuples {(1,1),(1,3)}
☐ A relation with scheme (A,C) and tuples {(1,1),(1,3),(2,6)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

*A relation with scheme (A,C) and tuples {(1,1),(1,3),(2,6)}*16) For R1, R2 to be *union-compatible* , they should have the same attribute names.**0.5 points**

- ☒ True
☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

*False*17) For R1, R2 to be *union-compatible* , they should have equal number of attributes.**0.5 points**

- ☒ True
☐ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

True

18) For R1, R2 to be *union-compatible* , the domains of the corresponding attributes should be the same.

0.5 points

☒ True

☐ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

True

19) The “divides” operator requires the operands to be *union-compatible* .

0.5 points

☐ True

☒ False

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

False